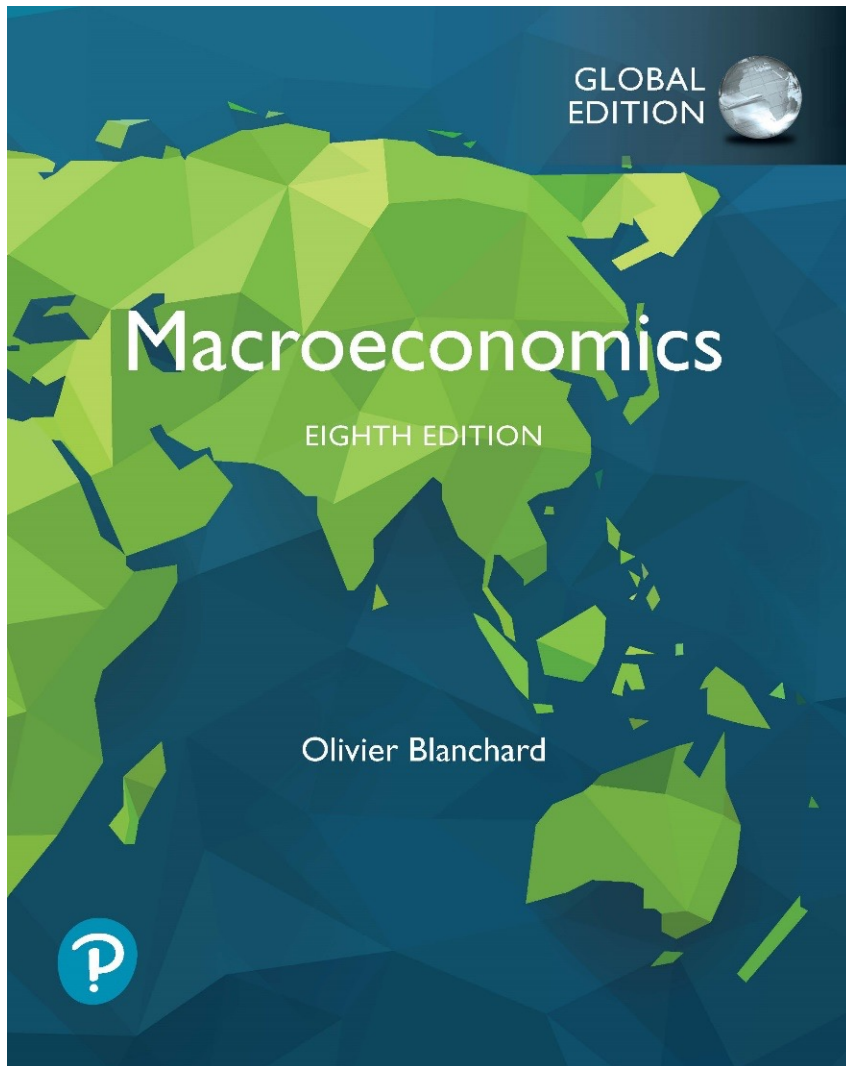


# Macroeconomics

Eighth Edition, Global Edition



## Chapter 9

From the Short to the Medium Run:  
The *IS-LM-PC* Model

# Chapter 9 Outline

## From the Short to the Medium Run: The *IS-LM-PC* Model

### 9.1 The IS-LM-PC Model

### 9.2 From the Short to the Medium Run

### 9.3 Complications and How Things Can Go Wrong

### 9.4 Fiscal Consolidation Revisited

### 9.5 The Effects of an Increase in the Price of Oil

### 9.6 Conclusions

# From the Short to the Medium Run: The *IS-LM-PC* Model

- In Chapters 3 through 6, we looked at equilibrium in the goods and financial markets, and how output is determined in the short run.
- In Chapters 7 and 8, we looked at equilibrium in the labor market.
- We now put the two parts together and use this IS-LM-PC (PC for Phillips curve) model to characterize the behavior of output both in the short run and the medium runs.

# 9.1 The IS-LM-PC Model (1 of 3)

- In Chapter 6, output in the short run is determined by demand (IS curve):

$$Y = C(Y - T) + I(Y, r + x) + G \quad (9.1)$$

- In Chapter 8, the relation between inflation and unemployment is called the Phillips curve:

$$\pi - \pi^e = -\alpha(u - u_n) \quad (9.2)$$

- When  $u = u_n$ , natural employment is  $N_n = L(1 - u_n)$
- When  $u = u_n$ , **potential output** is  $Y_n = L(1 - u_n)$
- Output gap:  $Y - Y_n = L((1 - u) - (1 - u_n)) = -L(u - u_n)$

$$Y - Y_n = -L(u - u_n) \Rightarrow \frac{Y - Y_n}{L} = -(u - u_n)$$

$$u = \frac{U}{U+N} = \frac{L-N}{L} = 1 - \frac{N}{L}$$

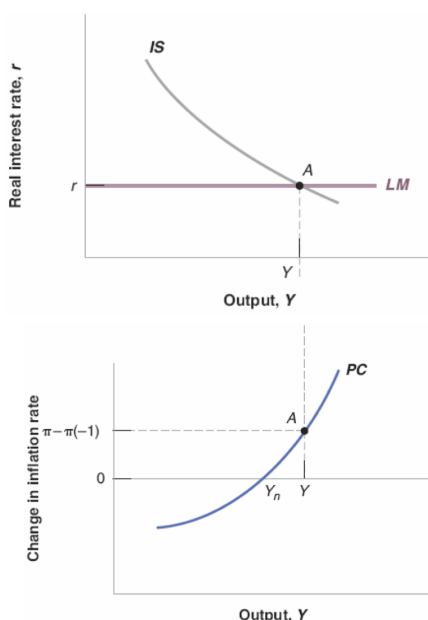
$$\frac{N}{L} = 1 - u$$

$$Y = L(1 - u) = N$$

$$\Rightarrow \underbrace{\pi - \pi^e}_{\pi^e = \pi_{t-1}} = -\alpha(u - u_n) = \underbrace{\frac{\alpha}{L}}_{Y=Y_n} (Y - Y_n)$$

$$\Rightarrow \pi = \pi^e + \frac{\alpha}{L} (Y - Y_n)$$

$$r \downarrow \rightarrow Y \uparrow \rightarrow \pi \uparrow (P \uparrow)$$



Good market: IS  $Y = C(Y - T) + I(Y, r + \lambda) + G$

Money market: LM  $\frac{M}{P} = Y \cdot L(\lambda) \quad r = \lambda - \pi^e$

|              | M      | B               | opp. cost of M                     |
|--------------|--------|-----------------|------------------------------------|
| $\pi = 0$    | 0      | $\lambda$       | $\lambda - 0 = \lambda$            |
| $\pi \neq 0$ | $-\pi$ | $\lambda - \pi$ | $\lambda - \pi - (-\pi) = \lambda$ |

Labor market:  $\frac{l}{1+m} = F(u, z)$  (ws and Ps eqm.)

$$u_n = \frac{m+z}{\alpha} \quad \text{if } F(u, z) = 1 + z - \alpha u; \quad Y_n = N_n = L(1 - u_n)$$

$$\Rightarrow PC: \pi - \pi^e = -\alpha(u - u_n) = \frac{\alpha}{L} (Y - Y_n)$$

In MR:  $\pi^e = \pi \Rightarrow Y = Y_n, N = N_n, U = U_n$   
(cont.)

$r = r_n$  it can be negative

$$\Rightarrow IS-LM-PC: r^*, Y^* (c^*, i^*)$$

In SR:  $\pi^e \neq \pi \Rightarrow Y \neq Y_n, N \neq N_n, U \neq U_n, r \neq r_n$   
Business cycle

$$p^* (P \neq p^e \rightarrow Y \neq Y_n)$$

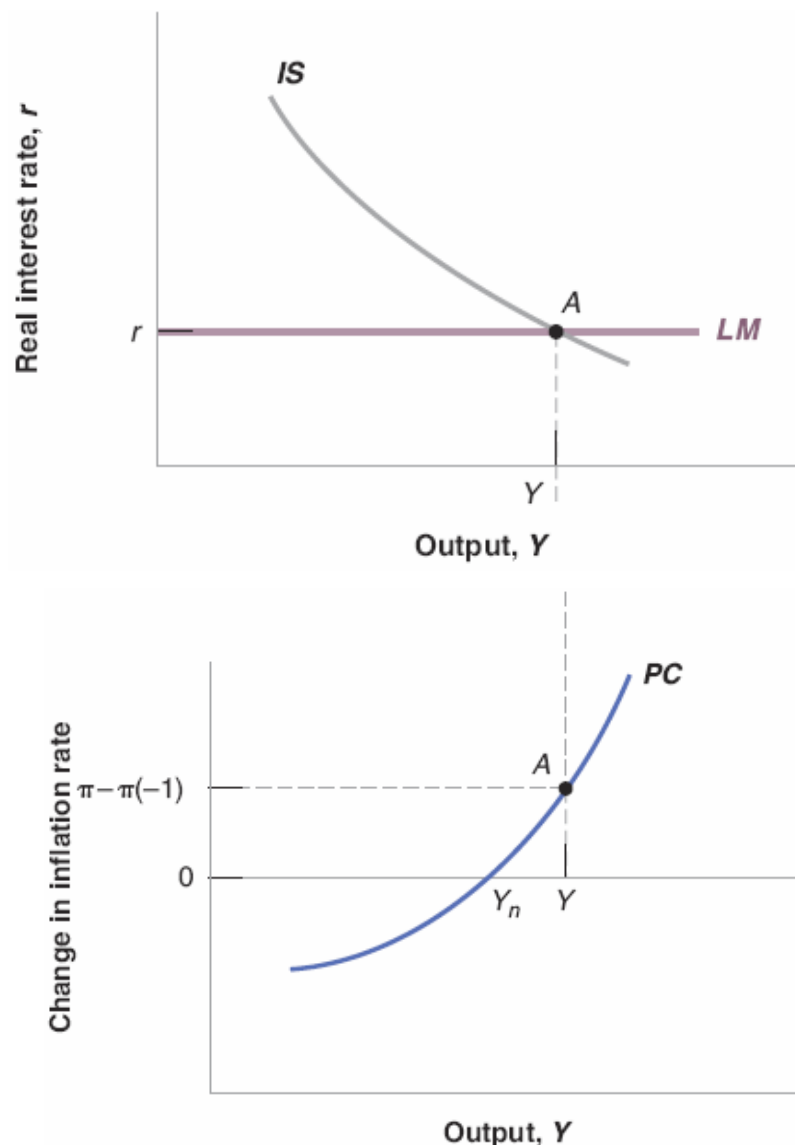
# 9.1 The IS-LM-PC Model (2 of 3)

## Figure 9.1 The IS-LM-PC Model: Output and Inflation

Short-run equilibrium

Top graph: A low policy rate leads to higher output.

Bottom graph: A higher output leads to a larger change in inflation.



# 9.1 The IS-LM-PC Model (3 of 3)

- Output gap:  $Y - Y_n = L((1 - u) - (1 - u_n)) = -L(u - u_n)$
- Replacing  $u - u_n$  in equation (9.2) gives:

$$\pi - \pi^e = (\alpha/L)(Y - Y_n) \quad (9.3)$$

- Assume wage setters expect inflation this year to be the same last year:

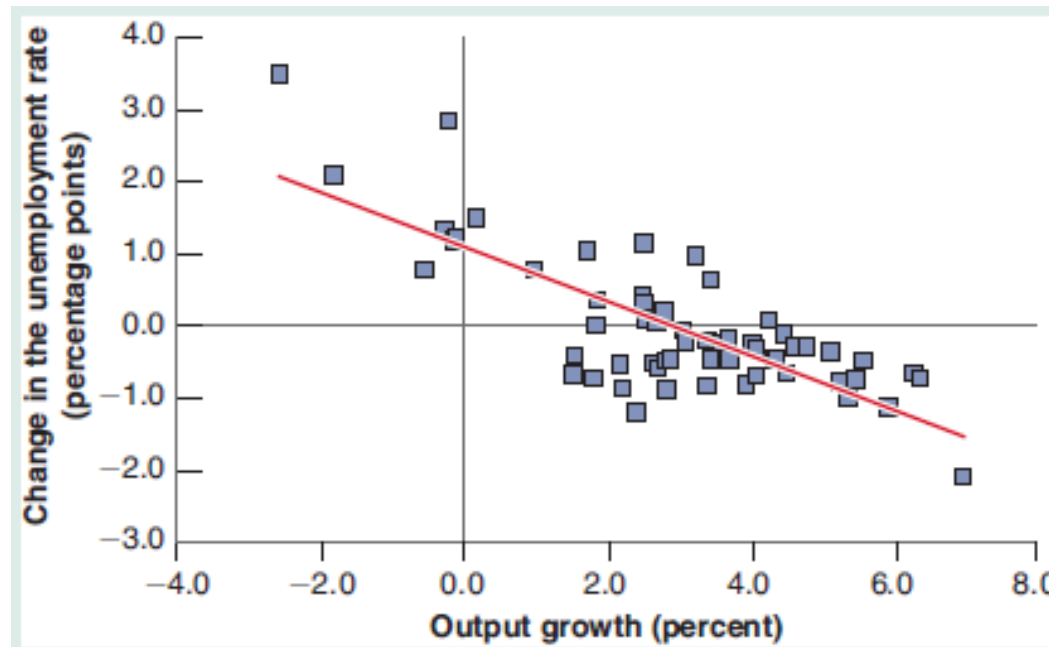
$$\pi - \pi(-1) = (\alpha/L)(Y - Y_n) \quad (9.4)$$

- When output is above potential (positive output gap), inflation increases, and vice versa.

# FOCUS: Okun's Law across Time and Countries (1 of 2)

**Figure 1** Changes in the Unemployment Rate versus Output Growth in the United States, 1960–2018

High output growth is associated with a reduction in the unemployment rate; low growth is associated with an increase in the unemployment rate.



Source: FRED: Series GDPCA, UNRATE



# FOCUS: Okun's Law across Time and Countries (2 of 2)

- Okun's law can be written as the change in the unemployment that is approximately equal to the negative of the growth rate of output:

$$u - u(-1) \approx -g_y \quad (9 \text{ B. } 1)$$

- The regression that fits the points in Figure 1 is:

$$u - u(-1) = -0.4(g_y - 3\%) \quad (9 \text{ B. } 2)$$

- Annual output growth has to be at least 3% to prevent the unemployment rate from rising.
- Output growth 1% above normal leads only to a 0.4% reduction in the unemployment rate due to such factors as labor hoarding and discouraged workers.
- The coefficient (0.4) is called the **Okun coefficient**.

# 9.2 From the Short to the Medium Run

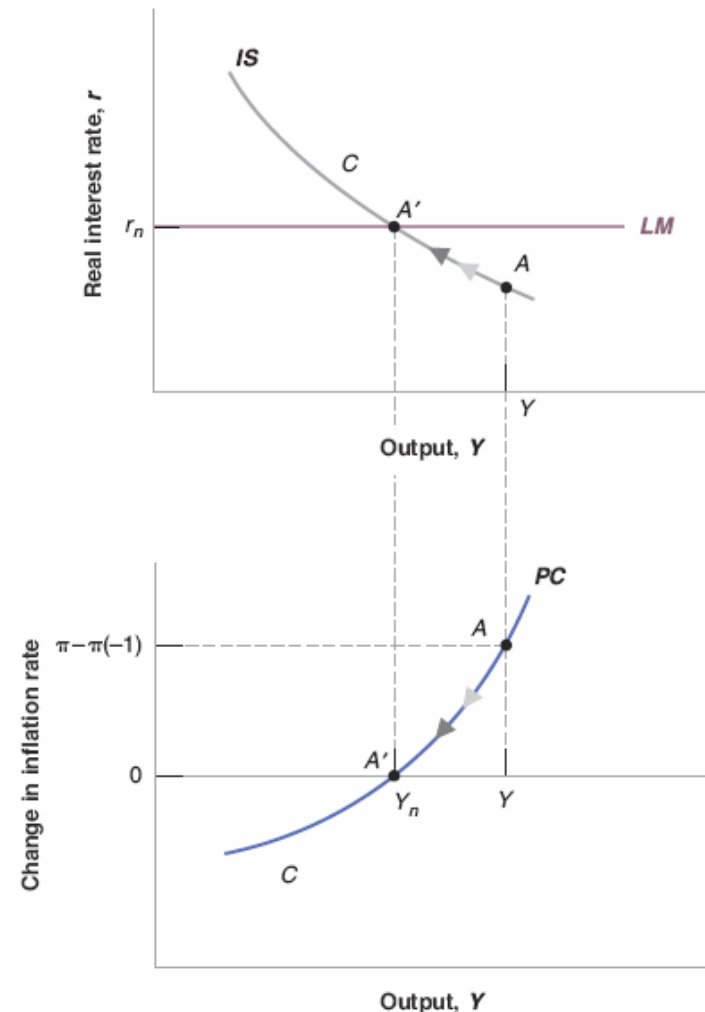
## Run (1 of 2)

**Figure 9.2** Medium-Run Output and Inflation

Over the medium run, the economy converges to the natural level of output and stable inflation.

At the medium-run equilibrium (point  $A'$ ),  $r_n$  is called the **natural, neutral, or Wicksellian rate of interest**.

If the central bank wants to achieve a constant level of inflation, then the initial boom must be followed by a recession.



## 9.2 From the Short to the Medium Run (2 of 2)

- In the medium run, the real variables are independent of monetary policy.
- Monetary policy determines the rate of inflation and the nominal rate of interest.
- The fact that monetary policy does not affect real variables in the medium run is referred to as the **neutrality of money**.

\* Transaction equation

$$M \cdot V = P \cdot T$$

$$\Rightarrow M \cdot V = P \cdot Y_N \quad (LR)$$

↓  
const.

$$\pi^e = (1-\theta)\bar{\pi} + \theta\pi_{t-1}$$

$$PC: \textcircled{1} \Delta\pi = \pi_t - \pi_{t-1} = \frac{\alpha}{L}(Y - Y_N)(\pi^e = \pi_{t-1})$$

$$\textcircled{2} \Delta\pi = \pi_t - \bar{\pi} = \frac{\alpha}{L}(Y - Y_N)(\pi^e = \bar{\pi})$$

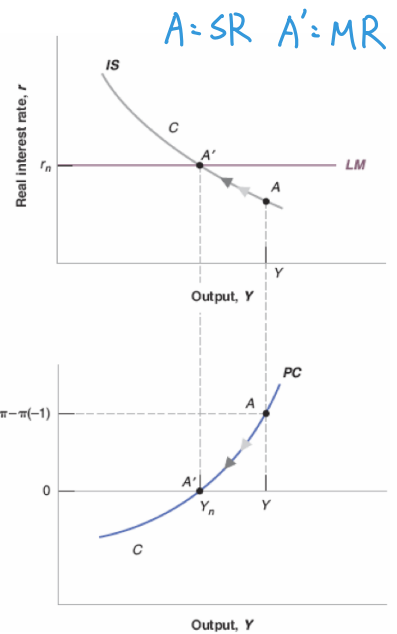
$$* Y > Y_N \Rightarrow \Delta\pi > 0, \pi \uparrow, P \uparrow \quad A \rightarrow A'$$

$$< \quad < \quad \downarrow \quad \downarrow \quad C \rightarrow A'$$

$$SR \quad > MR$$

$$\text{at } A \quad A' \Rightarrow P \uparrow (\pi \uparrow), r \uparrow, Y \downarrow$$

$$\text{at } C \quad A' \Rightarrow P \downarrow (\pi \downarrow), r \downarrow, Y \uparrow$$



\* 当  $\Delta\pi = 0$ , 但  $\pi$  太高  $\Rightarrow$  central bank 想要  $\pi \downarrow$ ,

只能使  $r \uparrow \quad A \rightarrow C$ ,  $\pi$  會比  $A \rightarrow A'$  要低, 但 economy 會進入 recession

inflation rate can be a prediction of output gap

$\textcircled{1} \Delta\pi > 0$  ( $P \uparrow$ ) increase  $r$  further  $\rightarrow$  recession

$r \uparrow, Y \downarrow$  ( $\pi = \text{const.}$ )

$\textcircled{2} \pi_t > 0 \quad A \rightarrow A' \text{ and } \pi_t \downarrow \rightarrow \text{recession}$

$r \uparrow, Y \downarrow$  (同時  $\pi_t \downarrow$ )  $\Rightarrow \pi = \bar{\pi} = \pi^e$

\* From SR to MR

$\textcircled{1} a. A \rightarrow A', \Delta\pi > 0, P \uparrow, r \uparrow, Y \downarrow \quad r = r_N, Y = Y_N, \pi = \text{const.}$

$\pi^e = \pi_{t-1}$  b. But in MR,  $\pi$  can be high. So need to move to C (Tight monetary policy) Recession

$\Rightarrow r > r_N, Y < Y_N$

$\Delta\pi < 0 \rightarrow P \downarrow, C \rightarrow A', r = r_N, Y = Y_N$ , But at lower const.  $\pi$

$\textcircled{2} A \rightarrow A', \pi \downarrow$  ( $P \downarrow$ ),  $r = r_N, Y = Y_N, \pi^e = \pi = \bar{\pi}$

$\pi^e = \bar{\pi}$

no policy cost.

$$\frac{M}{P} = Y L(\bar{\pi}) = \text{const.} \quad \frac{\Delta M}{M} = \frac{\Delta P}{P} = \pi$$

$$M^s \uparrow \Leftrightarrow i \downarrow \begin{cases} I \uparrow \rightarrow AD \uparrow \rightarrow Y \uparrow \\ (\frac{M}{P}) \uparrow \rightarrow \text{Income} \uparrow \rightarrow C \uparrow ; I \uparrow \rightarrow AD \uparrow \rightarrow Y \uparrow \end{cases} \Rightarrow Y \uparrow \text{ in SR}$$

$$\frac{M}{P} = Y \cdot L(i)$$

\* in SR

$$Y = C(Y - T) + I(Y, r) + G \uparrow$$

in LR

$$\bar{Y}_N = C(\bar{Y}_N - T) + I(\bar{Y}_N, r) + G \uparrow$$

$I \downarrow = G \uparrow$  for  $Y = Y_N$  completing crowd-out

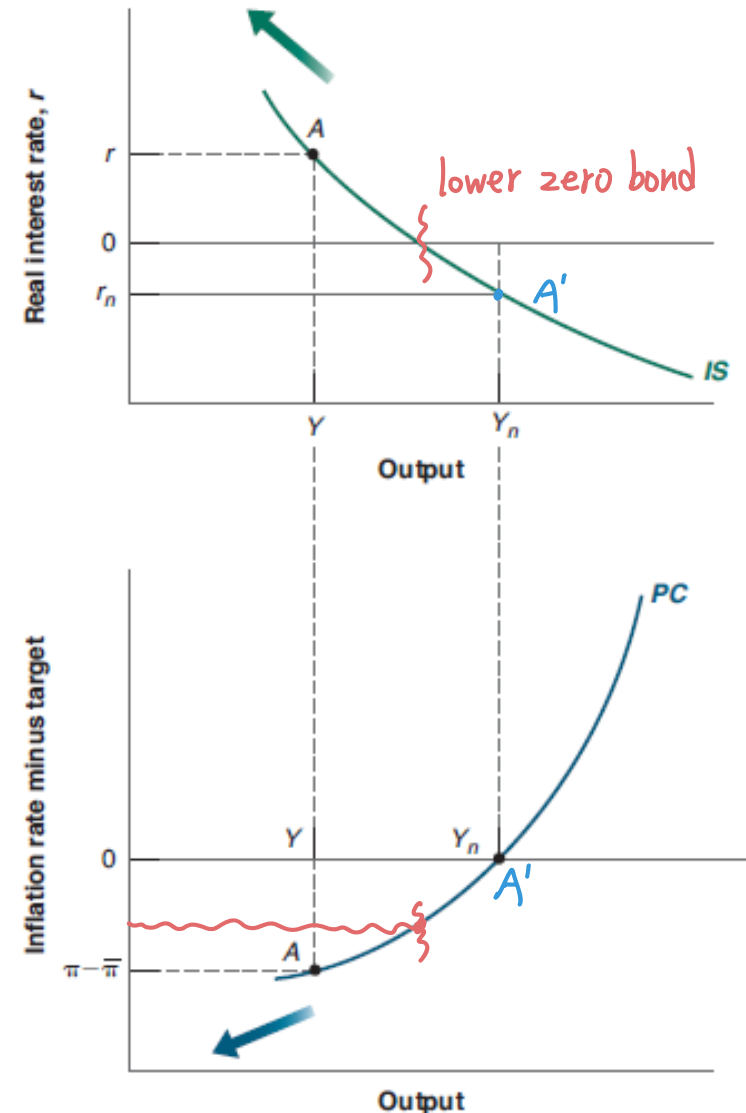
## 9.3 Complications and How Things Can Go Wrong (1 of 3)

- The adjustment to the medium-run equilibrium in Section 9-2 seemed smooth and easy.
- While this process looks easy to do in Figure 9-2, reality is more complicated for two reasons;
  1. It is difficult for the central bank to know where potential output is exactly and thus how far output is from potential.
  2. It takes time for the economy to respond.

# 9.3 Complications and How Things Can Go Wrong (2 of 3)

Figure 9.3 The Deflation Spiral

- If the zero lower bound prevents monetary policy from increasing output back to potential, the result may be a **deflation spiral**. More deflation leads to a higher real rate, which in turn leads to lower output and more deflation.



## 9.3 Complications and How Things Can Go Wrong (3 of 3)

- One issue that can occur is known as a deflation spiral.
- A **deflation spiral** or **deflation trap** results when deflation pushes real interest rates higher, lowers output, and leads to larger and larger deflation.  $r \uparrow, Y \downarrow, \pi \downarrow$
- This scenario played out during the Great Depression.
- The recent Great Recession gave rise to similar concerns.



# FOCUS: Deflation in the Great Depression

- The economy seemed to be in a deflation trap between 1929 and 1933.
- Monetary policy decreased the nominal interest rate from 5.3% to 2.3% in 1933, but with negative inflation rates, the real rate reached 12.3% in 1931 and 7.8% in 1933.

**Table 1** The Nominal Interest Rate, Inflation, and the Real Interest Rate, 1929–2033

| Year | Unemployment Rate (%) | Output Growth Rate (%) | One-Year Nominal Interest Rate (%), $i$ | Inflation Rate (%), $\pi$ | One-Year Real Interest Rate (%), $r$ |
|------|-----------------------|------------------------|-----------------------------------------|---------------------------|--------------------------------------|
| 1929 | 3.2                   | −9.8                   | 5.3                                     | 0.0                       | 5.3                                  |
| 1930 | 8.7                   | −7.6                   | 4.4                                     | −2.5                      | 6.9                                  |
| 1931 | 15.9                  | −14.7                  | 3.1                                     | −9.2                      | 12.3                                 |
| 1932 | 23.6                  | −1.8                   | 4.0                                     | −10.8                     | 14.8                                 |
| 1933 | 24.9                  | 9.1                    | 2.6                                     | −5.2                      | 7.8                                  |

# 9.4 Fiscal Consolidation Revisited

**Figure 9.4** Fiscal Consolidation in the Short and the Medium Run

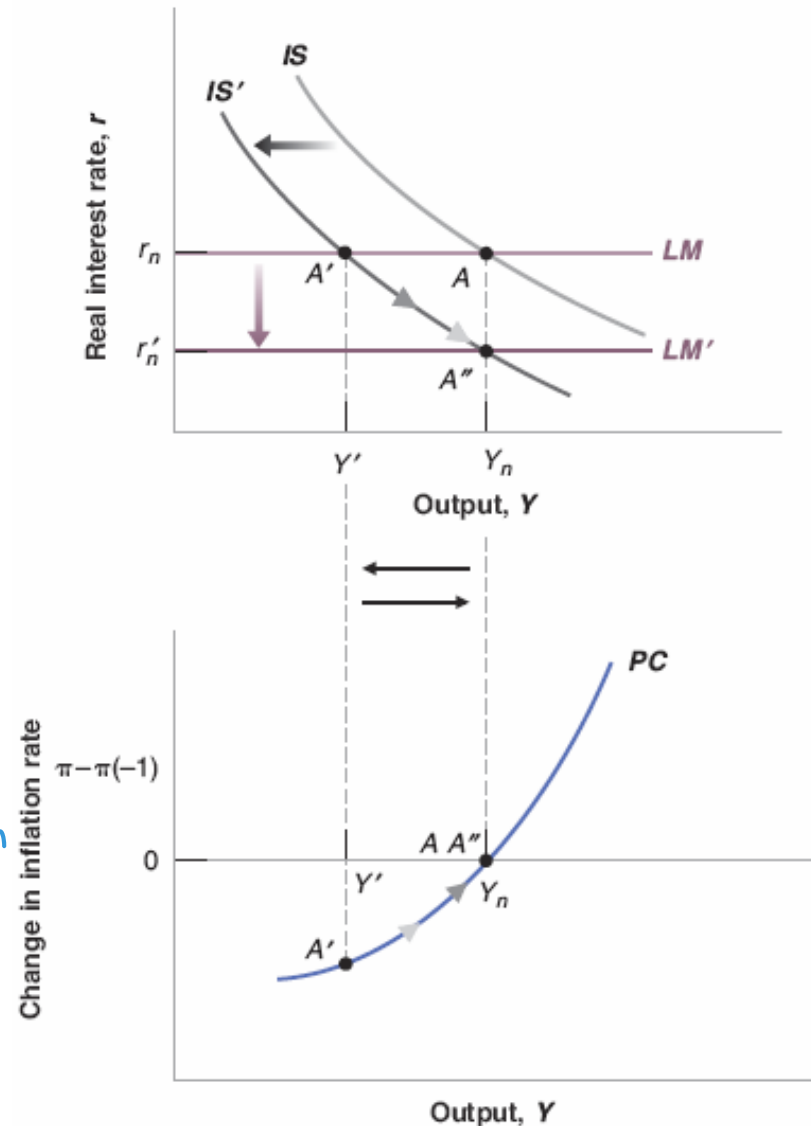
Fiscal consolidation leads to a decrease in output in the short run. In the medium run, output returns to potential, and the interest rate is lower.

$G \downarrow \Rightarrow IS$  line shifts left

in SR  $\Rightarrow Y \downarrow$  from  $A \rightarrow A' \Rightarrow C \downarrow, I \downarrow$   
 $\Rightarrow \underline{Y < Y_n}$  ( $\Delta \pi < 0$ )

in LR  $G \downarrow = I \uparrow \rightarrow K \uparrow \rightarrow Y \uparrow$  ( $P \downarrow$ )

$A'$  會逐漸回到  $A''$  透過 transitional dynamic mechanism



Accommodation by  $M^s \uparrow$   
 $(r) \downarrow$

from  $A' \rightarrow A''$  immediately

Expansional Fiscal Policy and Expansional Monetary Policy  
escape  $A'$  from  $A \rightarrow A''$

$Y_N$  will remain, but with lower  $r$ , without deflation

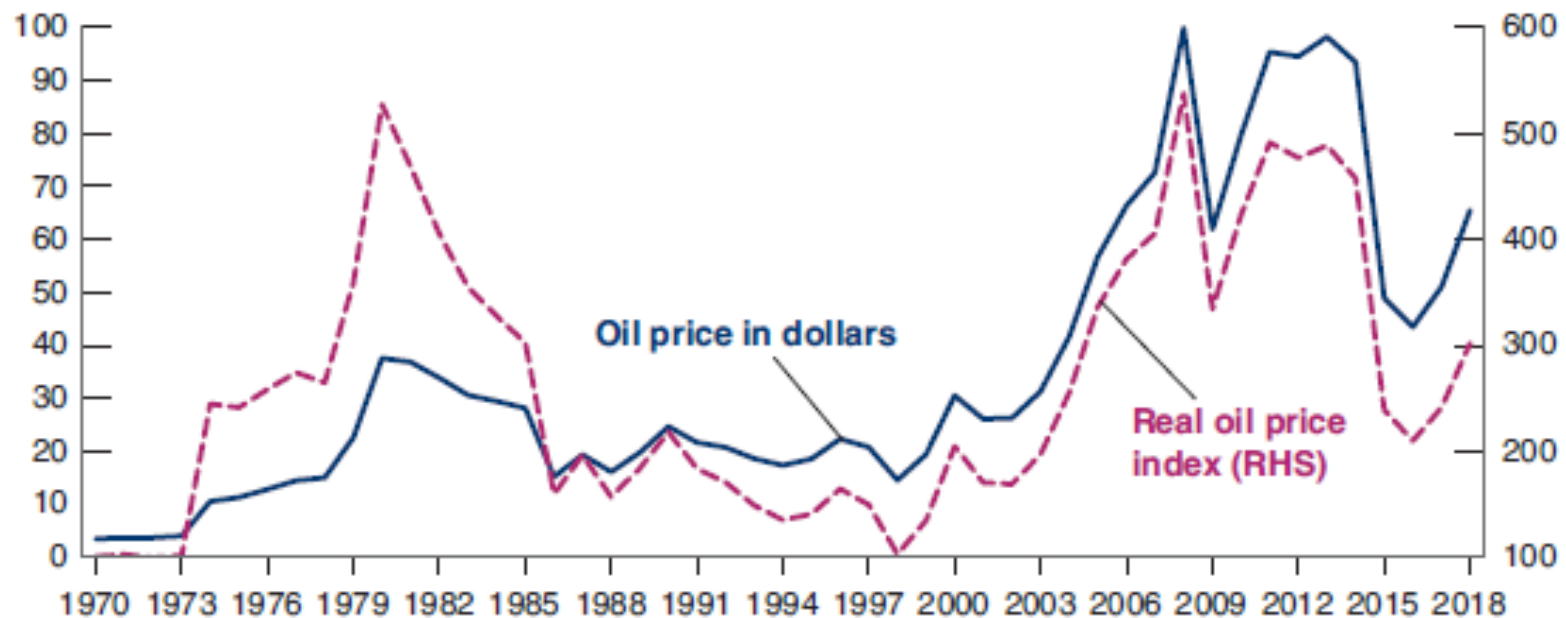
$C \downarrow = I \uparrow$ ,  $Y = Y_N$  same as before

Shorten the period with right policy combined.

# 9.5 The Effects of an Increase in the Price of Oil (1 of 4)

**Figure 9.5** The Nominal and the Real Price of Oil, 1970–2018

Over the last 40 years, there have been two sharp increases in the real price of oil, the first in the 1970s and the second in the 2000s.



- *Source:* FRED: OILPRICE, CPIAUSCL. Left vertical axis: nominal price of oil, dollars per barrel. Right vertical axis: real oil price index, 1970 = 100.

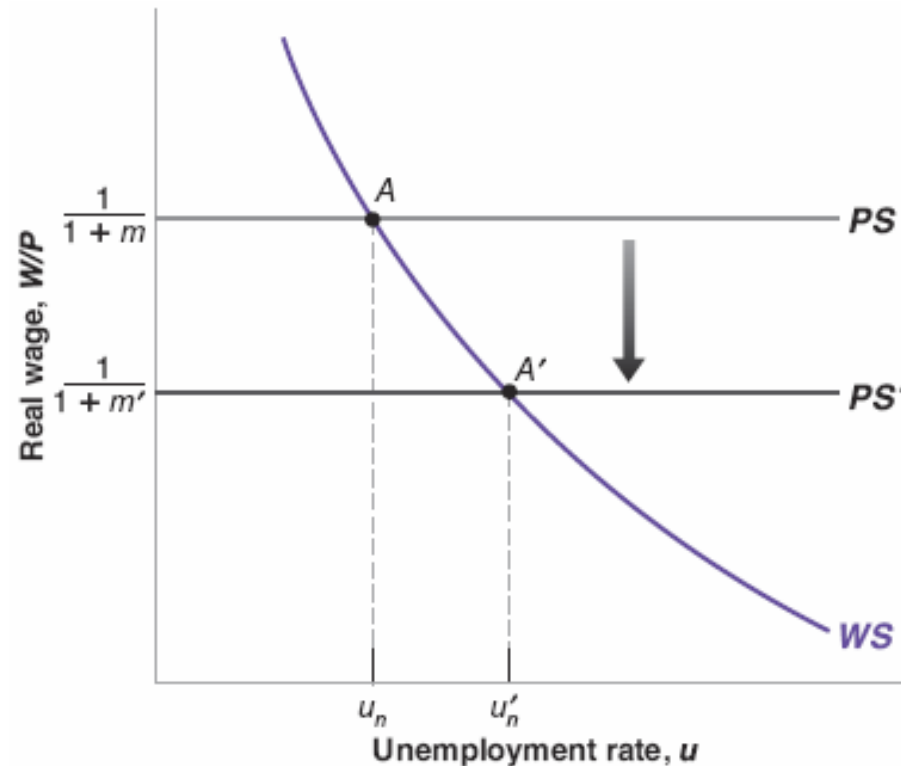
## 9.5 The Effects of an Increase in the Price of Oil (2 of 4)

- 1970s: OPEC (the Organization of Petroleum Exporting Countries) act as a monopoly and increased oil prices.
- 2000s: The fast growth of emerging economies led a rapid increase in world oil demand, and thus a steady increase in real oil prices.
- 2008: A large recession led to a sudden decrease in the demand for oil, and thus falling oil prices.
- 2014 and after: A combination of increased supply due to the increase in U.S. shale oil production and the partial breakdown of OPEC led to sudden drop in oil prices.

# 9.5 The Effects of an Increase in the Price of Oil (3 of 4)

**Figure 9.6** The Effects of an Increase in the Price of Oil on the Natural Rate of Unemployment

An increase in the price of oil is equivalent to an *increase in the markup*. It leads to lower real wages and a higher natural rate of unemployment.

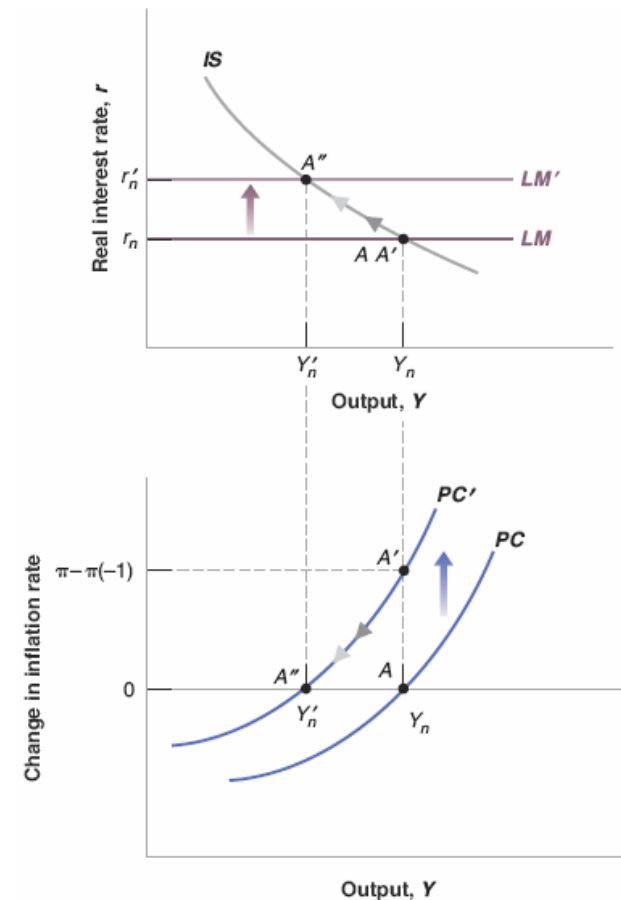


# 9.5 The Effects of an Increase in the Price of Oil (4 of 4)

**Figure 9.7** Short- and Medium-Run Effects of an Increase in the Price of Oil

- In the short run, an increase in the price of oil leads to higher inflation. If the price increase is permanent, it leads to lower output in the medium run.
- **Stagflation** (lower output and higher inflation) occurs along the way.

Supply shock

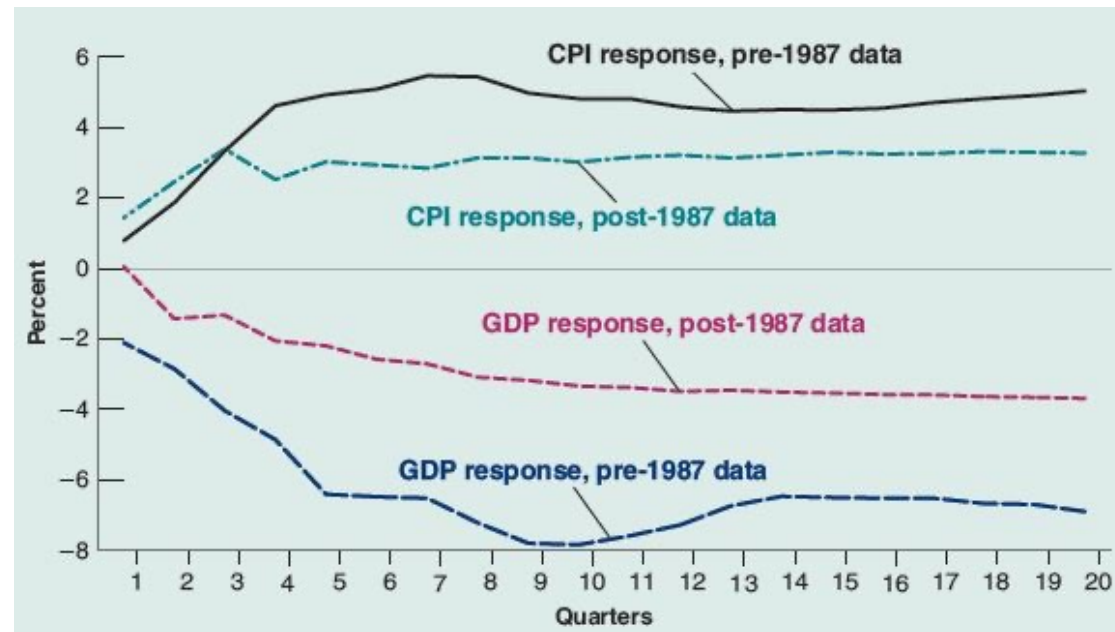


# FOCUS: Oil Price Increases: Why Were the 2000s So Different from the 1970s?

**Figure 1** The Effects of a 100% Permanent Increase in the Price of Oil on the CPI and on GDP

The effects of an increase in the price of oil on output and the price level are smaller than they used to be.

Plausible explanations include U.S. workers' lower bargaining power, and inflation that was more anchored by monetary policy.





## 9.6 Conclusions

- Shocks or changes in policy typically have different effects in the short run and in the medium run.
- Disagreements about the effects of various policies depend on how fast you think the economy adjusts to shocks.
- Movements in output around its trend are called **output fluctuations (business cycles)**.
- **Economic fluctuations** are the results of shocks and their dynamic effects, called the **propagation mechanism**.